

FINAL PROJECT PRESENTATION

<https://veggie-chef-81612c49.base44.app>

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1.CORE APPLICATION IDEA

Helping new cooks turn ingredients into confidence.

- Beginner-friendly
- Ingredient-based cooking
- Recipe matching
- Personalization

This app helps beginner cooks decide what to make based on the ingredients they already have. While following recipes is easy, choosing ingredients and figuring out what dishes are possible is often the hardest part. The app lets users select vegetables, meats, staples, and cooking tools, then instantly shows matching home-style recipes. It also supports custom recipe creation, allowing users to name and save their own combinations. This makes cooking more approachable, personalized, and fun—especially for those just starting to cook independently.

2.APP NAME

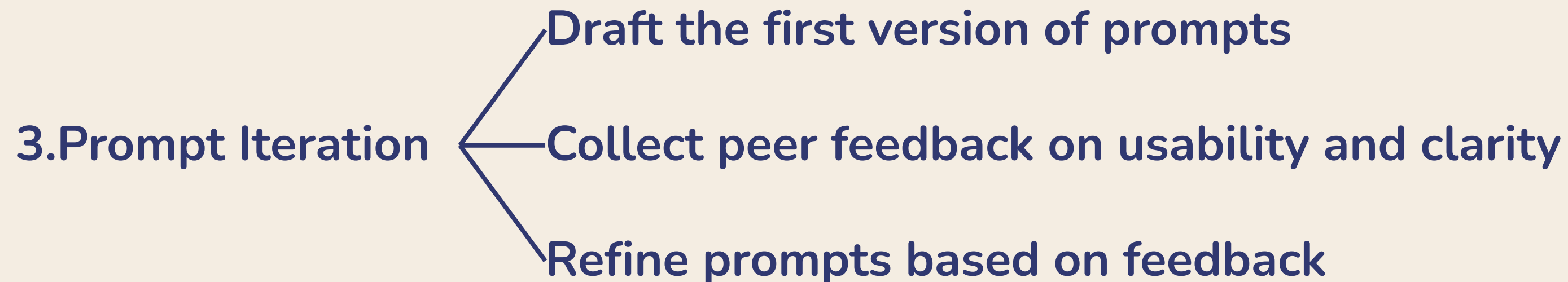
Recipe Matcher / 菜谱配对师

- Matching
- Guidance
- Personalization
- Confidence

3.TASK BREAKDOWN

1. Define app goals and core functions

2. Design main pages, visual style, and recipe-matching logic



4. Refine prompts based on feedback

5. Summarize challenges and reflect on learning

4.PROMPT STRUCTURE & ACTUAL PROMPT

1.Introduction and Goal

2.Page 1: Home

You are building a mobile app with a cute, clean, and vegetable-themed visual style. The app contains three main pages:

Page 1: Home – Ingredient & Tool Selection + Recipe Generation

The home screen is vertically structured into five sections:

1. Vegetable Selection Area

Includes: tomato, potato, carrot, cauliflower, onion, mushroom, eggplant, tofu, cabbage, cucumber

Each item is displayed as a tag with an image (see reference illustrations) and its name.

2. Meat Selection Area

Includes: chicken, pork, beef, luncheon meat, fish, shrimp, sausage, egg

Each item is displayed as a tag with an image and its name.

3. Staple Selection Area

Includes: rice, noodles, bread

Each item is displayed as a tag with an image and its name.

4. Cooking Tool Selection Area

Includes: Rice cooker, Oven, Microwave, Wok

Each tool is displayed as a tag with an image and its name.

5. Recipe Result Area

After the user selects at least one ingredient (vegetable, meat, or staple), and optionally a tool, the app matches the selection to the closest recipe(s) from the database below and displays the dish name(s).

3.Core Logic: Recipe Matching

Core Logic: Recipe Matching

Use the following recipe database to generate dish names based on user selections. Each entry follows the format:

[Ingredients + Tool → Dish Name]

Json

```
[
  { "combo": "Tomato + Egg + Rice cooker", "dish": "Tomato and Egg Rice Bowl" },
  { "combo": "Potato + Beef + Wok", "dish": "Stir-Fried Potato and Beef" },
  { "combo": "Carrot + Chicken + Rice cooker", "dish": "Carrot Chicken Stew Rice" },
  { "combo": "Cauliflower + Pork + Wok", "dish": "Cauliflower Pork Stir-Fry" },
  { "combo": "Onion + Sausage + Oven", "dish": "Baked Onion Sausage Casserole" },
  { "combo": "Mushroom + Egg + Microwave", "dish": "Microwaved Mushroom Egg Cup" },
  { "combo": "Eggplant + Beef + Rice cooker", "dish": "Braised Eggplant with Beef Ri
  { "combo": "Tofu + Shrimp + Wok", "dish": "Tofu and Shrimp Stir-Fry" },
  { "combo": "Cabbage + Pork + Wok", "dish": "Cabbage Pork Stir-Fry" },
  { "combo": "Cucumber + Chicken + Microwave", "dish": "Microwaved Chicken Cucumber
  { "combo": "Tomato + Luncheon meat + Rice cooker", "dish": "Tomato Luncheon Meat R
  { "combo": "Potato + Egg + Oven", "dish": "Baked Potato Egg Frittata" },
  { "combo": "Carrot + Beef + Wok", "dish": "Carrot Beef Stir-Fry" },
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  { "combo": "Eggplant + Sausage + Oven", "dish": "Baked Eggplant Sausage Tray" },
  { "combo": "Tofu + Egg + Microwave", "dish": "Microwaved Tofu Egg Bowl" },
  { "combo": "Cabbage + Beef + Wok", "dish": "Cabbage Beef Stir-Fry" },
  { "combo": "Cucumber + Fish + Rice cooker", "dish": "Fish Cucumber Rice Bowl" },
  { "combo": "Tomato + Egg + Bread + Oven", "dish": "Tomato Egg Breakfast Toast" },
  { "combo": "Potato + Chicken + Rice cooker", "dish": "Chicken Potato Rice Stew" },
  { "combo": "Carrot + Pork + Wok", "dish": "Carrot Pork Stir-Fry" },
  { "combo": "Cauliflower + Egg + Microwave", "dish": "Microwaved Cauliflower Egg Mi
  { "combo": "Onion + Beef + Wok", "dish": "Onion Beef Stir-Fry" },
  { "combo": "Mushroom + Shrimp + Rice cooker", "dish": "Mushroom Shrimp Rice Bowl"
  { "combo": "Eggplant + Chicken + Wok", "dish": "Eggplant Chicken Stir-Fry" },
  { "combo": "Tofu + Fish + Oven", "dish": "Baked Tofu Fish Tray" },
  { "combo": "Cabbage + Egg + Microwave", "dish": "Microwaved Cabbage Egg Bowl" },
  { "combo": "Cucumber + Pork + Wok", "dish": "Cucumber Pork Stir-Fry" }
```

复制

Matching logic:

User must select at least one ingredient (vegetable, meat, or staple)

Tool selection is optional but improves accuracy

Match user input to the closest recipe(s) and display the dish name(s)

4.Page 2: Custom Recipe Creation

5.Page 3: Favorites

6.Visual Style Requirements

Page 2: Custom Recipe Creation

- Users can create their own recipe by selecting ingredients and a tool from the same ingredient/tool library.
- They manually input a custom dish name.
- This new recipe is added to the recipe database.
- When the same combination is selected again, the custom dish name will appear in the results.

Page 3: Favorites

- After a successful recipe match on Page 1, a ⭐ icon appears next to the dish name.
- Users can tap the star to save the combination and dish name to their personal favorites.
- Page 3 displays all saved combinations and dish names.

Visual Style Requirements

- The app should use a **cute, clean, and vegetable-themed illustration style**.
- All ingredient and tool tags must include both **image and name**.
- Use the provided cartoon-style assets as visual references for vegetables, meats, staples, and tools.

5. PEER FEEDBACKS

1. Expand recipe variety

The current recipe database is clear but limited. Adding more diverse dishes, including vegetarian options or quick snacks, would make the app more appealing to different users.

2. Enhance user interaction

Instead of only showing dish names, the app could display short descriptions or images of the dishes. This would make the results more engaging and help beginners visualize what they are cooking.

6.PROMPT REFINEMENT

1.Visual Reference

To improve clarity and visual consistency, I added picture references at the end of the prompt, covering vegetables, meats, staples, and tools. This helped guide the app’s visual generation more effectively.

2.Scalability

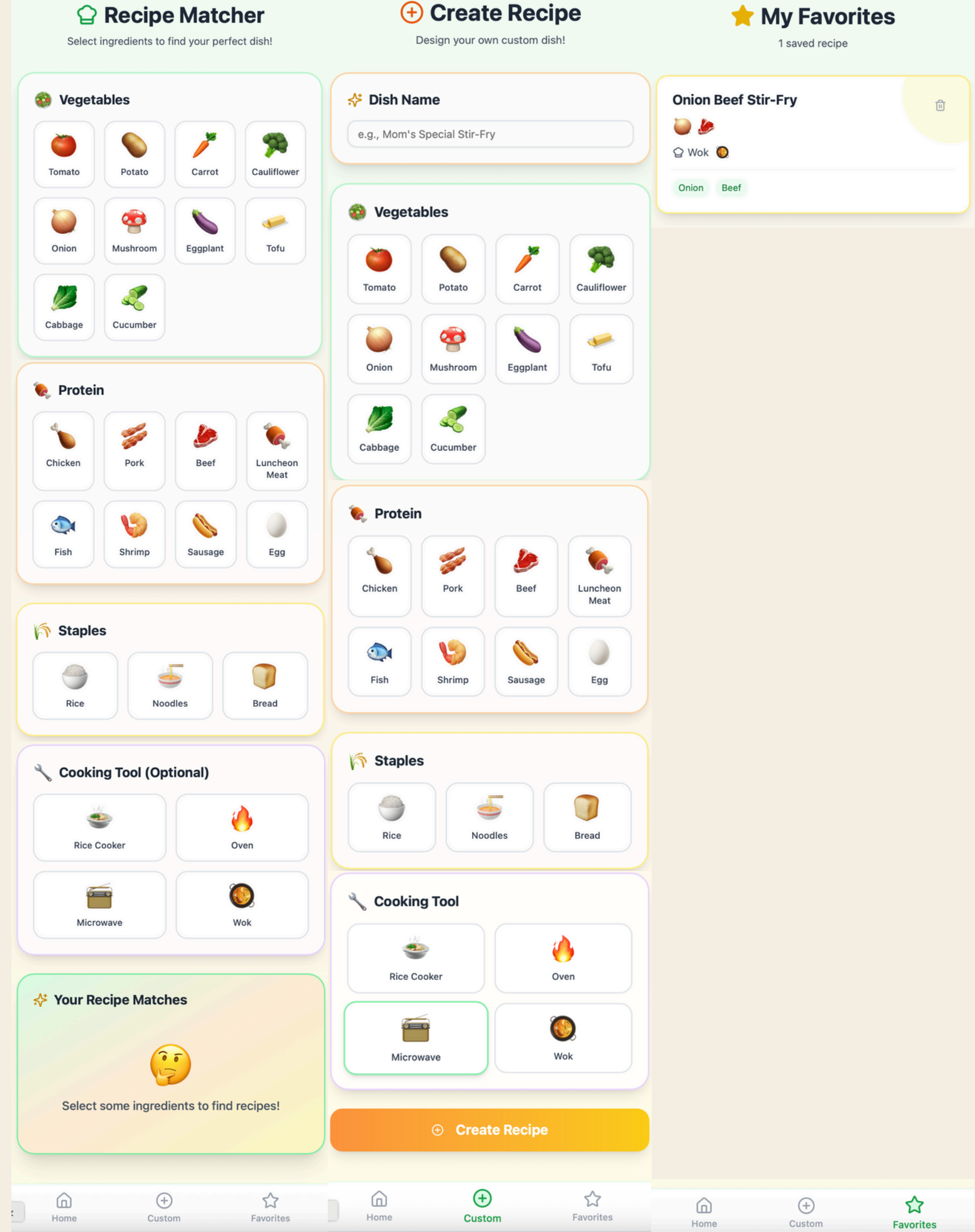
Regarding recipe variety, I decided not to expand the database at this stage, since the app is currently for demonstration. Adding new recipes is simple—just follow the same format and update the database. This modular design makes future expansion easy and flexible.

meats, staples, and tools. (picture
references:vegetables:https://copilot.microsoft.com/1103-4c66-bdba-347584f9104c.png;meat::https://copilot.microsoft.com/6c91-41bd-968e-beb32677cece.png;staple
foods:https://copilot.microsoft.com/th/id/BCC012da-412b-9687-b9e8f5b75a08.png;cookers:blob:https://copilot.microsoft.com/77ee-41b6-9f79-16577dabb41a/blob:https://copilot.microsoft.com/0a0c-4496-b374-c23dba1e050e/blob:https://copilot.microsoft.com/c883-4ca0-b9d6-298072d85acc

Json ^ 复制

```
[
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  { "combo": "Potato + Beef + Wok", "dish": "Stir-Fried Potato and Beef" },
  { "combo": "Carrot + Chicken + Rice cooker", "dish": "Carrot Chicken Stew Rice" },
  { "combo": "Cauliflower + Pork + Wok", "dish": "Cauliflower Pork Stir-Fry" },
  { "combo": "Onion + Sausage + Oven", "dish": "Baked Onion Sausage Casserole" },
  { "combo": "Mushroom + Egg + Microwave", "dish": "Microwaved Mushroom Egg Cup" },
  { "combo": "Eggplant + Beef + Rice cooker", "dish": "Braised Eggplant with Beef Rice" },
  { "combo": "Tofu + Shrimp + Wok", "dish": "Tofu and Shrimp Stir-Fry" },
  { "combo": "Cabbage + Pork + Wok", "dish": "Cabbage Pork Stir-Fry" },
  { "combo": "Cucumber + Chicken + Microwave", "dish": "Microwaved Chicken Cucumber" },
  { "combo": "Tomato + Luncheon meat + Rice cooker", "dish": "Tomato Luncheon Meat Rice Bowl" },
  { "combo": "Potato + Egg + Oven", "dish": "Baked Potato Egg Frittata" },
  { "combo": "Carrot + Beef + Wok", "dish": "Carrot Beef Stir-Fry" },
  { "combo": "Cauliflower + Shrimp + Wok", "dish": "Cauliflower Shrimp Stir-Fry" },
  { "combo": "Onion + Chicken + Rice cooker", "dish": "Onion Chicken Rice Stew" },
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  { "combo": "Eggplant + Sausage + Oven", "dish": "Baked Eggplant Sausage Tray" },
  { "combo": "Tofu + Egg + Microwave", "dish": "Microwaved Tofu Egg Bowl" },
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  { "combo": "Cucumber + Fish + Rice cooker", "dish": "Fish Cucumber Rice Bowl" },
  { "combo": "Tomato + Egg + Bread + Oven", "dish": "Tomato Egg Breakfast Toast" },
  { "combo": "Potato + Chicken + Rice cooker", "dish": "Chicken Potato Rice Stew" },
  { "combo": "Carrot + Pork + Wok", "dish": "Carrot Pork Stir-Fry" },
  { "combo": "Cauliflower + Egg + Microwave", "dish": "Microwaved Cauliflower Egg Muffin" },
  { "combo": "Onion + Beef + Wok", "dish": "Onion Beef Stir-Fry" },
  { "combo": "Mushroom + Shrimp + Rice cooker", "dish": "Mushroom Shrimp Rice Bowl" },
  { "combo": "Eggplant + Chicken + Wok", "dish": "Eggplant Chicken Stir-Fry" },
  { "combo": "Tofu + Fish + Oven", "dish": "Baked Tofu Fish Tray" },
  { "combo": "Cabbage + Egg + Microwave", "dish": "Microwaved Cabbage Egg Bowl" },
  { "combo": "Cucumber + Pork + Wok", "dish": "Cucumber Pork Stir-Fry" }
```

7.FINAL OUTPUT



7.FINAL OUTPUT

(1)Home Page



Recipe Matcher

Select ingredients to find your perfect dish!



Vegetables



Tomato



Potato



Carrot



Cauliflower



Onion



Mushroom



Eggplant



Tofu



Cabbage



Cucumber



Protein



Chicken



Pork



Beef



Luncheon Meat



Fish



Shrimp



Sausage



Egg



Staples



Rice



Noodles



Bread



Cooking Tool (Optional)



Rice Cooker



Oven



Microwave



Wok



Your Recipe Matches



Select some ingredients to find recipes!



Home



Custom



Favorites

(2) Custom Recipe Page

+ Create Recipe

Design your own custom dish!

Dish Name

e.g., Mom's Special Stir-Fry

Vegetables

Tomato

Potato

Carrot

Cauliflower

Onion

Mushroom

Eggplant

Tofu

Cabbage

Cucumber

The image displays a grid of food items categorized into two groups: Protein and Staples. The Protein section is at the top, featuring a header with a chicken drumstick icon and the word "Protein". Below the header are eight items arranged in two rows of four: Chicken (chicken drumstick), Pork (two strips of pork), Beef (a slice of beef), Luncheon Meat (a slice of luncheon meat), Fish (a blue fish), Shrimp (two shrimp), Sausage (two sausages), and Egg (a white egg). The Staples section is at the bottom, featuring a header with a bowl of rice icon and the word "Staples". Below the header are three items arranged in a single row: Rice (a bowl of white rice), Noodles (a bowl of orange noodles), and Bread (a slice of white bread). Each item is represented by a realistic illustration within a rounded square frame.

Protein			
			
Chicken	Pork	Beef	Luncheon Meat
			
Fish	Shrimp	Sausage	Egg

Staples		
		
Rice	Noodles	Bread



Cooking Tool



Rice Cooker



Oven



Microwave



Wok



Create Recipe



Home



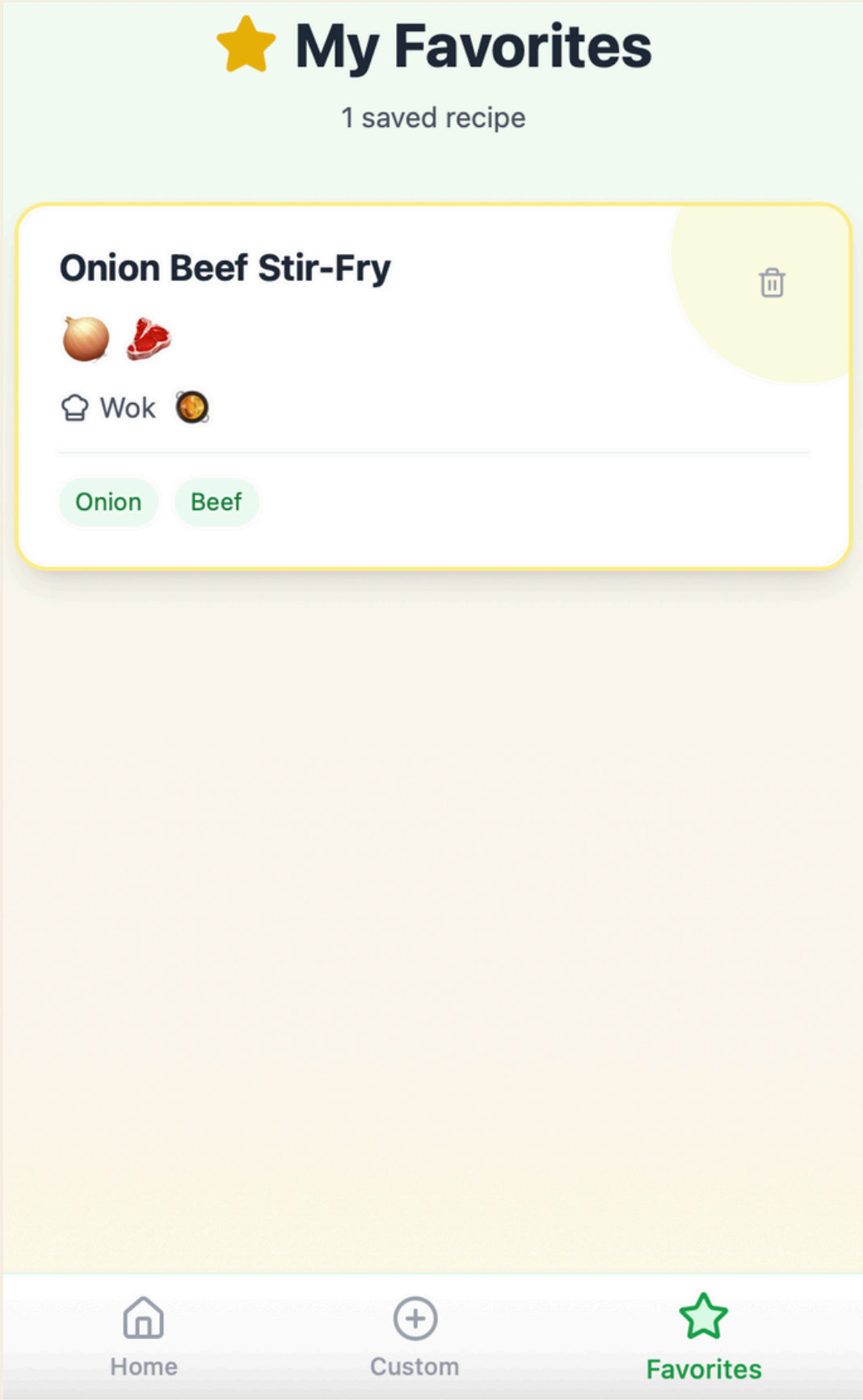
Custom



Favorites

7.FINAL OUTPUT

(3) Favorites Page



8. CHALLENGES

● AI Limitation

I couldn't integrate an AI model like OpenAI's API, which limited intelligent recipe generation. To solve this, I used a rule-based matching system with a local database—still customizable, but not AI-driven.

● Prompt Complexity

The prompt structure was complex. It had to cover layout, logic, and visual style clearly and consistently, which required careful planning.

● Function vs. Clarity

I had to prioritize clarity over full functionality, keeping the recipe database simple for demonstration purposes.

9. REFLECTIONS

Technical Constraints

This project challenged me to balance creativity with technical constraints. Although I couldn't integrate an AI model, I adapted by using rule-based matching and a local database to simulate intelligent behavior.

Modular Design

Peer feedback reminded me of the importance of variety and engagement, which I addressed through modular design and visual refinement.

Prompt Clarity

Structuring the prompt was demanding, as it had to cover layout, logic, and style clearly.

Peer Inspiration

Most importantly, I learned to listen to peers' suggestions and, by offering feedback myself, gained new inspiration to further refine my own ideas.

THANK YOU

Thanks For Watching

10. ITERATION (AFTER PRESENTATION)

New function:
Add Custom Ingredients

Create Recipe

Design your own custom dish!

Dish Name

e.g., Mom's Special Stir-Fry

+ Add Custom Ingredient

Vegetables

Tomato

Potato

Carrot

Cauliflower

Onion

Mushroom

Eggplant

Tofu

Home

Custom

Favorites

Add New Ingredient

Ingredient Name

e.g., Spinach, Bacon, Quinoa...

Category

Select a category

Choose an Icon

Tap to select emoji

+ Add Ingredient

<https://veggie-chef-81612c49.base44.app>